

TING-WEI LI

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I am interested in **data-centric machine learning**. I work on data-centric ML and the full data pipeline for modern AI systems, currently focusing on *data selection*, *data attribution*, and *data generation* for LLMs and agentic systems. I have authored 15+ papers at top venues including ICML/NeurIPS/ICLR, with 250+ citations.

EDUCATION

- **University of Illinois Urbana-Champaign** Champaign, IL
Ph.D. in Computer Science 08/2024 - Present
Advisor: Prof. [Hanghang Tong](#)
GPA: 4.0/4.0; Expected graduation: 12/2028
- **University of Michigan, Ann Arbor** Ann Arbor, MI
M.Sc. in Electrical and Computer Engineering 08/2021 - 04/2023
GPA: 4.0/4.0
- **National Taiwan University** Taipei, Taiwan
B.S.Eng in Electrical Engineering 09/2016 - 06/2020
GPA: 3.9/4.3

EXPERIENCE

- **Meta Platforms, Inc.**
Part-Time Student Researcher 08/2026 - Present, Remote, IL
 - Developed efficient data selection methods tailored for large-scale Ads recommendation models.
 - Mentors: Amit Anand Amlesahwaram, Bill Huang**Research Intern** 05/2026 - 08/2026, Sunnyvale, CA
 - Developed gradient-based data attribution methods tailored for large-scale Ads recommendation models.
 - Mentors: Amit Anand Amlesahwaram, Bill Huang
- **Amazon.com, Inc.**
Applied Scientist Intern 05/2025 - 08/2025, Seattle, WA
 - Designed an efficient multi-task learning framework for a large-scale video platform.
 - Mentors: Cody Severinski, Changhe Yuan
- **AT&T Inc.**
Chief Data Office Intern (Research Intern) 09/2023 - 11/2023, Bedminster, NJ
 - Developed interpretable graph-based models for large-scale customer behavior analysis.
 - Mentors: Qiong Wu, Zhengyi Zhou, Eric Zavesky

SELECTED PUBLICATIONS

1. **Ting-Wei Li**, Yuanchen Bei, Xiao Lin, Hanghang Tong. “*Beyond LLM-Based Reasoning: Lightweight GNNs for Agent Failure Attribution*”. In **Conference on Neural Information Processing Systems (NeurIPS)** 2026. [\[Paper\]](#).
2. **Ting-Wei Li**, Sirui Chen, Jiaru Zou, Yingbing Huang, Tianxin Wei, Jingrui He, Hanghang Tong. “*EvoSelect: Data-Efficient LLM Evolution for Targeted Task Adaptation*”. In **Conference on Language Modeling (COLM)** 2026. [\[Paper\]](#).

3. Tianxin Wei*, **Ting-Wei Li***, Zhining Liu*, Xuying Ning, Ze Yang, Jiaru Zou, Zhichen Zeng, Ruizhong Qiu, Xiao Lin, Dongqi Fu, Zihao Li, Mengting Ai, Duo Zhou, Wenxuan Bao, Yunzhe Li, Gaotang Li, Cheng Qian, Yu Wang, Xiangru Tang, Yin Xiao, Liri Fang, Hui Liu, Xianfeng Tang, Yuji Zhang, Chi Wang, Jiaxuan You, Heng Ji, Hanghang Tong, Jingrui He. “*A Survey of Agentic Reasoning for Large Language Models: Towards Recursively Self-Improving and Collective Agents*”. In **Transactions on Machine Learning Research (TMLR)** 2026. [Paper].
4. Xuying Ning, Dongqi Fu, Tianxin Wei, Mengting Ai, Jiaru Zou, **Ting-Wei Li**, Hanghang Tong, Yada Zhu, Hendrik Hamann, Jingrui He. “*MC-Search: Evaluating and Enhancing Multimodal Agentic Search with Structured Long Reasoning Chains*”. In **International Conference on Learning Representations (ICLR)** Oral 2026. [Paper].
5. Ruizhong Qiu*, **Ting-Wei Li***, Gaotang Li, Hanghang Tong. “*Graph Homophily Booster: Rethinking the Role of Discrete Features on Heterophilic Graphs*”. In **International Conference on Learning Representations (ICLR)** 2026. [Paper].
6. Hyunsik Yoo, **Ting-Wei Li**, SeongKu Kang, Zhining Liu, Charlie Xu, Qilin Qi, Hanghang Tong. “*Continual Low-Rank Adapters for LLM-based Generative Recommender Systems*”. In **International Conference on Learning Representations (ICLR)** 2026. [Paper].
7. **Ting-Wei Li**, Ruizhong Qiu, Hanghang Tong. “*Graph Data Selection for Domain Adaptation: A Model-Free Approach*”. In **Conference on Neural Information Processing Systems (NeurIPS)** 2025. [Paper] [Poster].
8. Junwei Deng*, **Ting-Wei Li***, Shiyuan Zhang, Shixuan Liu, Yijun Pan, Hao Huang, Xinhe Wang, Pingbang Hu, Xingjian Zhang, Jiaqi W Ma. “*dattri: A Library for Efficient Data Attribution*”. In **Conference on Neural Information Processing Systems (NeurIPS) Spotlight** 2024. [Paper] [Code] [Poster].
9. Junwei Deng*, **Ting-Wei Li***, Shichang Zhang, Jiaqi Ma. “*Efficient Ensembles Improve Training Data Attribution*”. In **Transactions on Machine Learning Research (TMLR)** 2024. [Paper].
10. **Ting-Wei Li**, Qiaozhu Mei, Jiaqi Ma. “*A Metadata-Driven Approach to Understand Graph Neural Networks*”. In **Conference on Neural Information Processing Systems (NeurIPS)** 2023. [Paper] [Code] [Poster].
11. Jiaqi Ma, Xingjian Zhang, Hezheng Fan, Jin Huang, Tianyue Li, **Ting-Wei Li**, Yiwen Tu, Chenshu Zhu, Qiaozhu Mei. “*Graph learning indexer: A contributor-friendly and metadata-rich platform for graph learning benchmarks*”. In **Learning on Graphs Conference (LoG)** 2022. [Paper] [Code].

(* indicates equal contribution)

PREPRINTS AND SUBMISSIONS

1. **Ting-Wei Li**, Ernie Chang, Yuanchen Bei, Narine Kokhlikyan, Hanghang Tong. “*HyperBridge: Effectively Adapting Hypernetworks across LLM Architectures*”. In **Arxiv (In submission to ICLR)** 2026.
2. Xinrui He, **Ting-Wei Li**, Junting Wang, Mengting Ai, Xinyu He, Hanghang Tong, Jingrui He. “*Harnessing Image Question Dependence for Better VLM Test-time Reinforcement Learning*”. In **Arxiv (In submission to ICLR)** 2026. [Paper].
3. Ruizhong Qiu*, **Ting-Wei Li***, Gaotang Li*, Tianxin Wei, Jingrui He, Hanghang Tong. “*Efficient Inference Scaling for Safety Assurance*”. In **Arxiv** 2025. [Paper].
4. Junwei Deng*, Yuzheng Hu*, Pingbang Hu*, **Ting-Wei Li***, Shixuan Liu*, Jiachen T. Wang, Dan Ley, Qirun Dai, Benhao Huang, Jin Huang, Cathy Jiao, Hoang Anh Just, Yijun Pan, Jingyan Shen, Yiwen Tu, Weiwei Wang, Xinhe Wang, Shichang Zhang, Shiyuan Zhang, Ruoxi Jia, Himabindu Lakkaraju, Hao Peng, Weijing Tang, Chenyan Xiong, Jieyu Zhao, Hanghang Tong, Han Zhao, Jiaqi W. Ma. “*A Survey of Data Attribution: Methods, Applications, and Evaluation in the Era of Generative AI*”. In **SSRN (In submission to JAIR)** 2025. [Paper].

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SKILLS

- **Programming:** Python, C++, Bash

- **ML & LLM Frameworks:** PyTorch, Hugging Face, vLLM, TRL, verl, PEFT
- **AI-Native Development:** Claude Code, Claude CLI, Cursor

SELECTED AWARDS & HONORS

- **Scholar Award**, Conference on Neural Information Processing Systems 2023
- **Graduate Student Research Assistant Tuition Waiver**, University of Michigan 2023

INVITED TALKS

- **Agentic Reasoning for Large Language Models**, NLP Academic Exchange Platform 2026
- **Data Attribution for GenAI Copyright**, Data Attribution Reading Group 2024

ACADEMIC SERVICES

- **Conference Reviewer**
 - ACM Web Conference (WWW) 2027
 - IEEE/CVF International Conference on Computer Vision (ICCV) 2025
 - Conference on Neural Information Processing Systems (NeurIPS) 2025, 2026
 - International Conference on Learning Representations (ICLR) 2025, 2026, 2027
 - International Conference on Machine Learning (ICML) 2024, 2025, 2026
 - IEEE International Conference on Big Data (Big Data) 2023
 - IEEE International Conference on Data Mining Workshop on Machine Learning on Graphs (ICDM MLoG) 2023
- **Program Committee**
 - AAAI Conference on Artificial Intelligence (AAAI) 2025, 2026, 2027
 - IEEE International Conference on Data Science and Advanced Analytics (DSAA) 2024
- **Organizer Team**
 - Data Foundations of AI Seminar Coordinator
 - Data Attribution Reading Group Speaker, Organization Member

TEACHING EXPERIENCE

- University of Illinois Urbana-Champaign, **CS 412: Introduction to Data Mining** Spring 2026